The Polynomial Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: OPERATE, GCF, FACTOR and SOLVE (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (OPERATE)
- ☐ **A2** (OPERATE)
- ☐ **A3** (GCF)
- ☐ **A4** (FACTOR)
- ☐ **A5** (FACTOR)
- ☐ **A6** (SOLVE)

Entrées

10 pts each · choose 4

- ☐ **E1** (OPERATE)
- ☐ **E2** (GCF)
- ☐ **E3** (FACTOR)
- ☐ **E4** (SOLVE)
- ☐ **E5** (OPERATE)
- ☐ **E6** (SOLVE)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (OPERATE)
- ☐ **S2** (FACTOR)
- ☐ **S3** (SOLVE)
- ☐ **S4** (GCF)

My skill check (circle each skill you have covered): OPERATE · GCF · FACTOR · SOLVE

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 OPERATE ☐ My pick · 5 pts

Add: $(3x^2 + 5x - 2) + (x^2 - 7x + 6)$

A2 OPERATE ☐ My pick · 5 pts

Multiply: $(x + 4)(x - 3)$

A3 GCF ☐ My pick · 5 pts

Factor out the greatest common factor: $12x^3 + 18x^2$

A4 FACTOR ☐ My pick · 5 pts

Factor: $x^2 + 7x + 12$

A5 FACTOR ☐ My pick · 5 pts

Factor: $4x^2 - 25$

A6 SOLVE ☐ My pick · 5 pts

Solve by factoring: $x^2 - 5x - 14 = 0$

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 OPERATE ☐ My pick · 10 pts

Multiply and simplify: $(2x - 3)(x^2 + 4x - 5)$

E2 GCF ☐ My pick · 10 pts

Factor completely: $3x^3 - 12x^2 - 36x$

E3 FACTOR ☐ My pick · 10 pts

Factor: $6x^2 + 11x - 10$. Show how you check your answer by multiplying.

E4 SOLVE ☐ My pick · 10 pts

Solve: $2x^2 + 5x = 12$

E5 OPERATE ☐ My pick · 10 pts

Lena subtracted:

$(5x^2 + 2x - 1) - (3x^2 - 4x + 6) = 2x^2 - 2x + 5$

Find her mistake, explain it, and give the correct answer.

E6 SOLVE ☐ My pick · 10 pts

Solve each by factoring:

(a) $3x^2 - 27 = 0$ **(b)** $x^2 + 6x + 9 = 0$ (Why does (b) give only one solution?)

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 OPERATE ☐ My pick · 20 pts

A rectangular pool is (2x + 3) m long and (x + 4) m wide. A deck 1 m wide surrounds the pool on all four sides. **(a)** Write the pool’s area as a polynomial. **(b)** Write the area of pool plus deck. **(c)** Find the deck’s area. **(d)** Evaluate (c) for x = 4.

S2 FACTOR ☐ My pick · 20 pts

Mia factored $4x^2 - 36$ as $(2x - 6)(2x + 6)$ and says she is done. **(a)** Show her answer multiplies back correctly. **(b)** Explain why it is not factored completely, and factor completely. **(c)** Factor completely: $2x^3 - 50x$.

S3 SOLVE ☐ My pick · 20 pts

A ball is thrown upward from an 80-foot cliff. Its height is $h = -16t^2 + 64t + 80$ feet after t seconds. **(a)** Factor the right side completely. **(b)** When does the ball hit the ground (h = 0)? Explain why one answer is rejected. **(c)** Besides t = 0, when is the ball again at 80 feet?

S4 GCF ☐ My pick · 20 pts

A rectangle has area $6x^3 + 15x^2 - 9x$ square units. Its width is the GCF of the three terms. **(a)** Find the width and the length. **(b)** Factor the length to write the area as a product of three factors. **(c)** Check with x = 2.